

Engineering Data Science

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Previously...

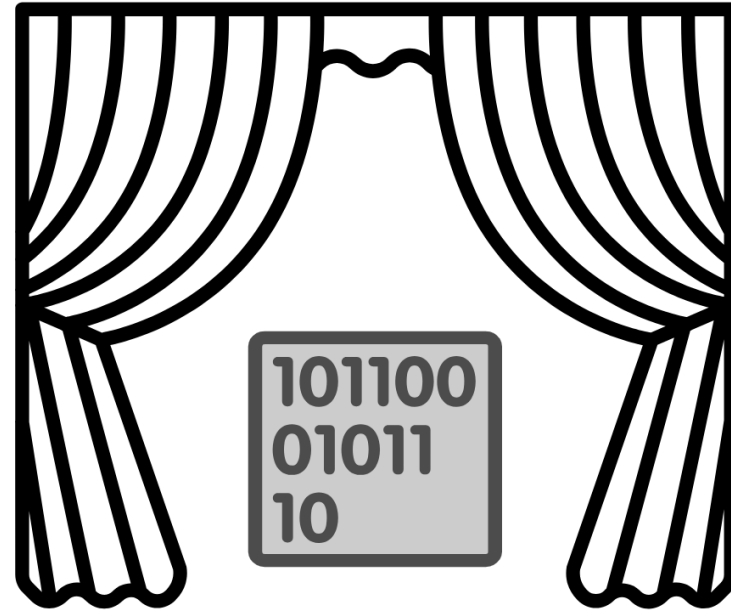
- Data Science Challenges

Previously...

- Data Science Challenges

- **Bias**

Systematic tendency in which methods used to gather data and compute statistics generate inaccurate depictions of reality.



Source: <https://mlatcl.github.io/advds/lectures/04-02-ai-and-data-science.html>

Challenges our ability to deploy safe and effective solutions:

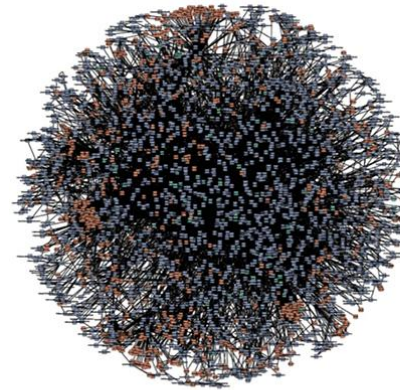
- **Alignment**
- **Fairness**
- **Inclusiveness**

Previously...

- Data Science Challenges

- **Complexity**

Systems are highly dynamic and have grown in size. The data processing pipelines involve hundreds or thousands of components.



amazon.com



NETFLIX

Source: <https://www.divante.com/blog/10-companies-that-implemented-the-microservice-architecture-and-paved-the-way-for-others>

Challenges our technical ability to deploy and maintain our solutions:

- **Sustainability**
- **Maintainability**

Previously...

- Data Science Challenges

- **Intellectual Debt**

Black-box components make systems hard to understand and threaten human control. We know they components work but do not know how.



Source: (Zittrain-2019)
<https://medium.com/berkman-klein-center/from-technical-debt-to-intellectual-debt-in-ai-e05ac56a502c>

Challenges our ability to explain our solutions:

- **Interpretability**
- **Accountability**

Why are these important?

- Data Science Challenges
 - Bias
 - Complexity
 - Intellectual Debt
 - ...



Source: <https://www.freepik.com/free-photos-vectors/society>

Why are these important?

- Society has challenging problems...



How do we build solutions now?

- Focus on technology



How do we build solutions now?

- Focus on technology

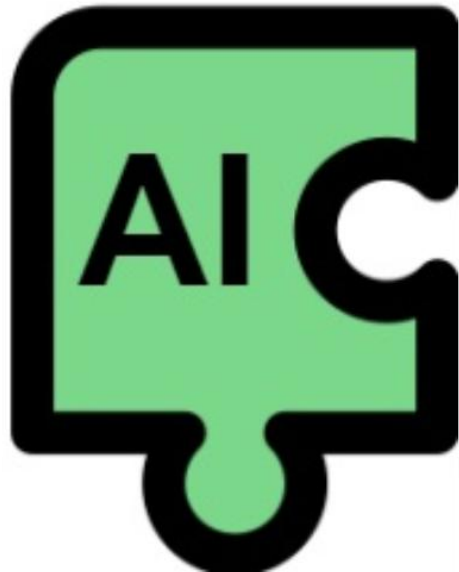


For example, we have a learning model that generates predictions from inputs:



How do we build solutions now?

- Focus on technology



For example, we have a learning model that generates predictions from inputs:



Learning Model	Accuracy
Model v1	88%
Model v2	90%
Model v3	98%

More capable models
Larger models
Powerful infrastructure
Promising applications

How do we build solutions now?

- Focus on technology



For example, we have a learning model that generates predictions from inputs:

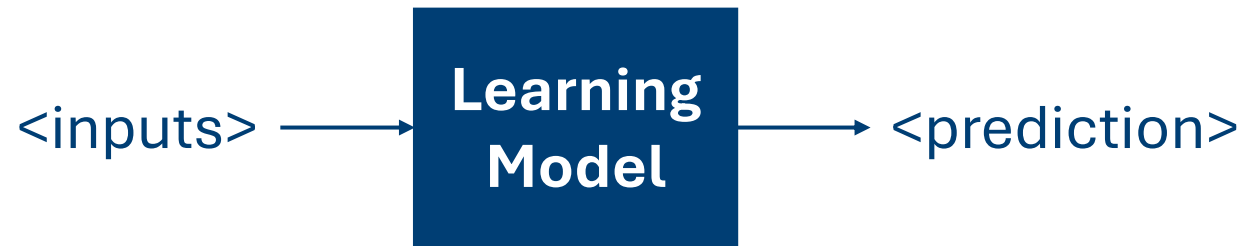


Learning Model	Accuracy
Model v1	88%
Model v2	90%
Model v3	98%

Does Model v3 address end users' requirements?

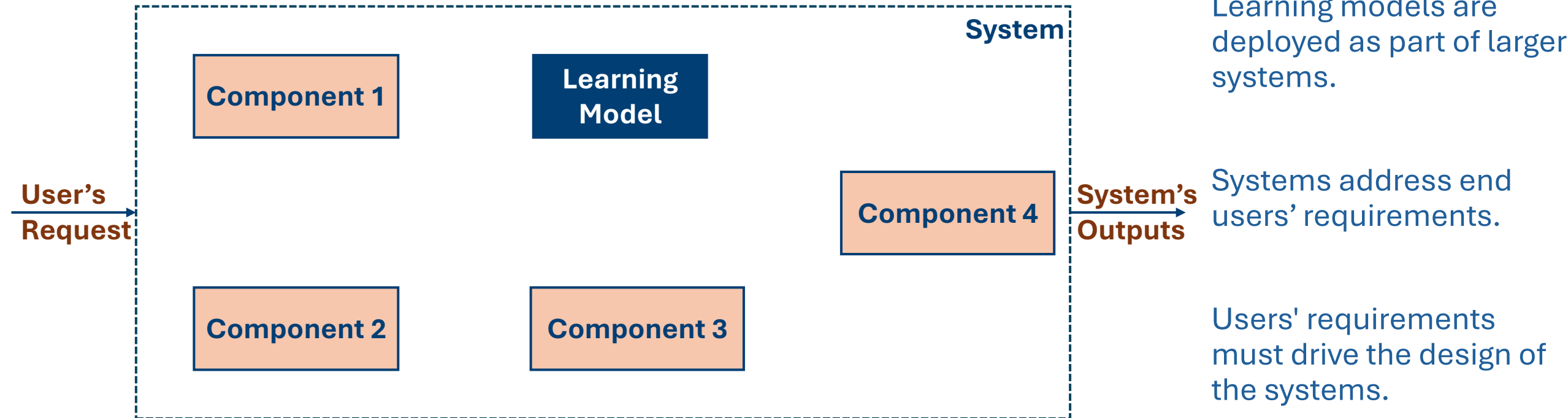
Real-world Deployments

- Context matters



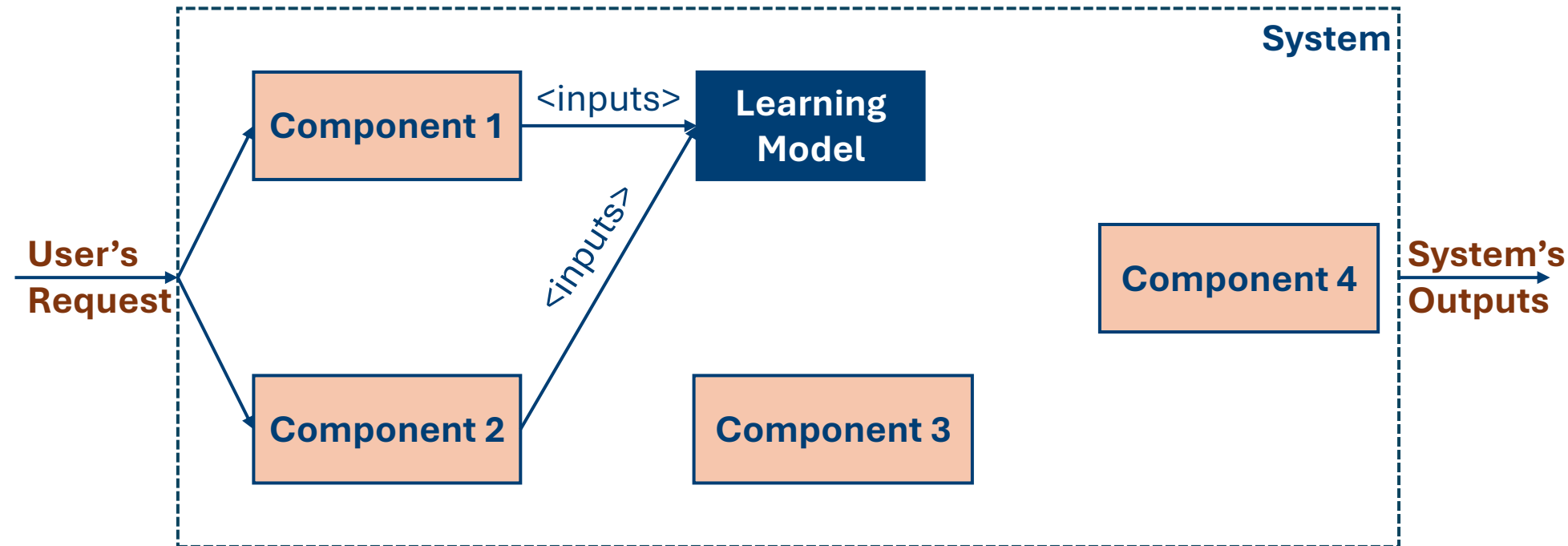
Real-world Deployments

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Real-world Deployments

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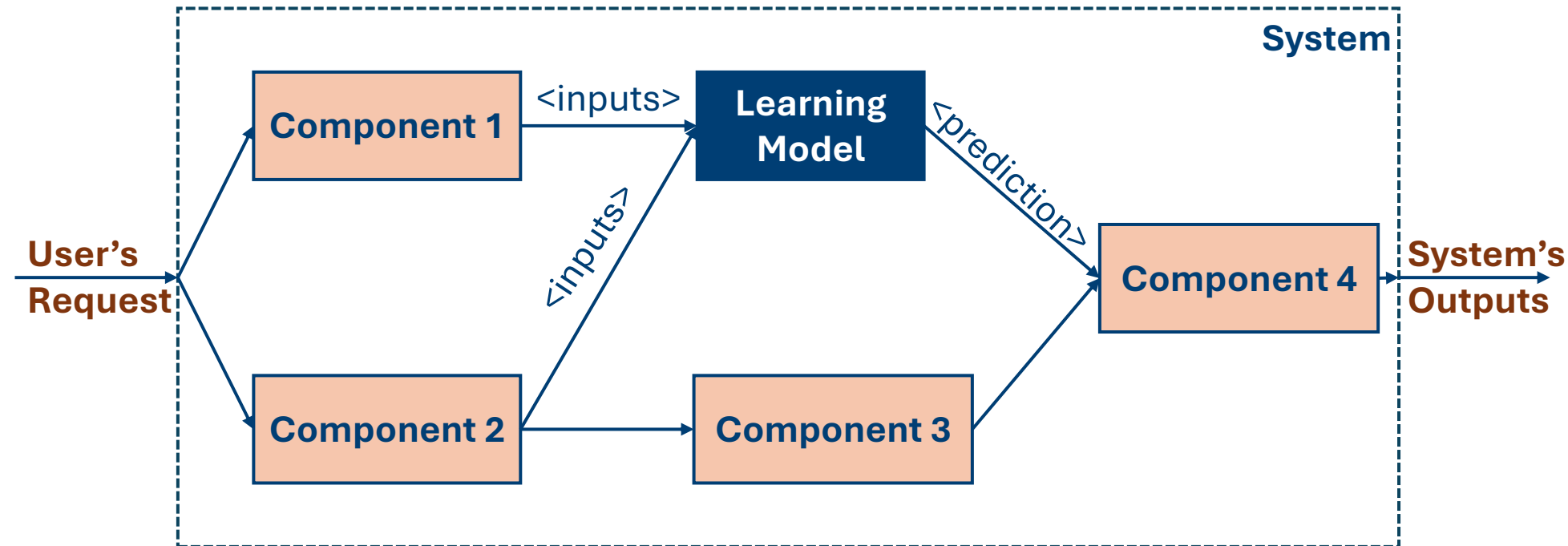


Systems are composed of multiple components

These components can impact the behavior of the learning models

Real-world Deployments

- Context matters

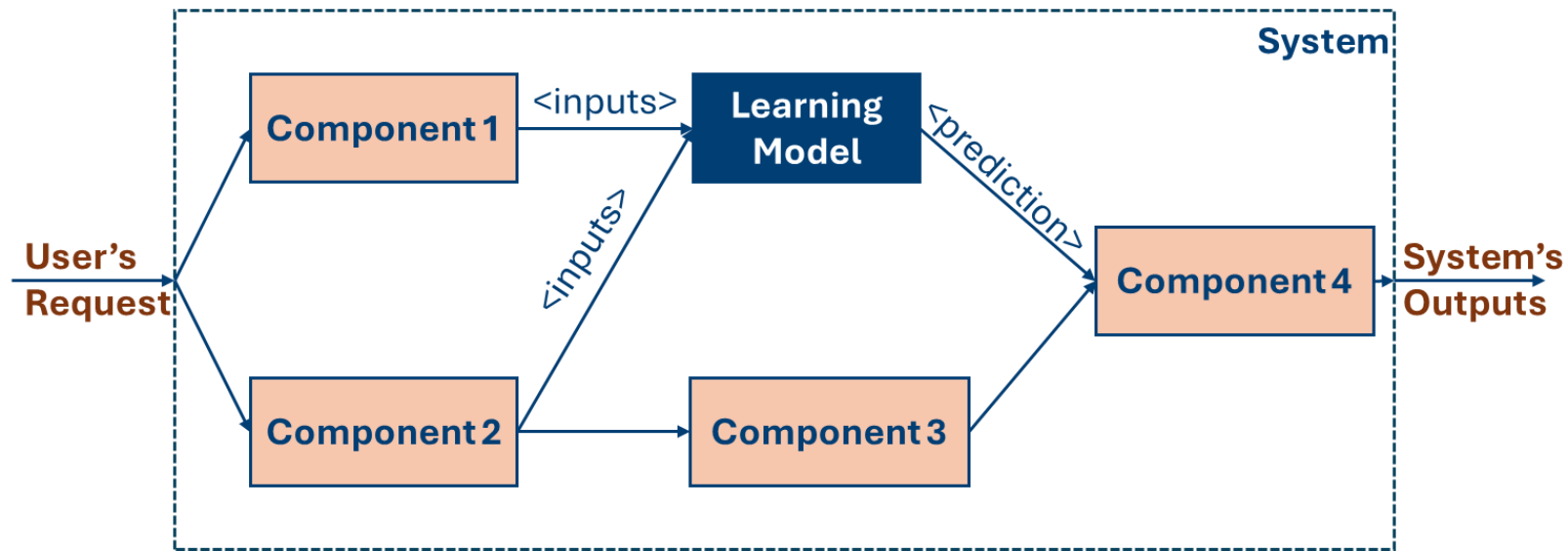


Learning models outputs can be inputs for other components

Systems' outputs are aggregations of the data processed by different components.

Real-world Deployments

- Context matters

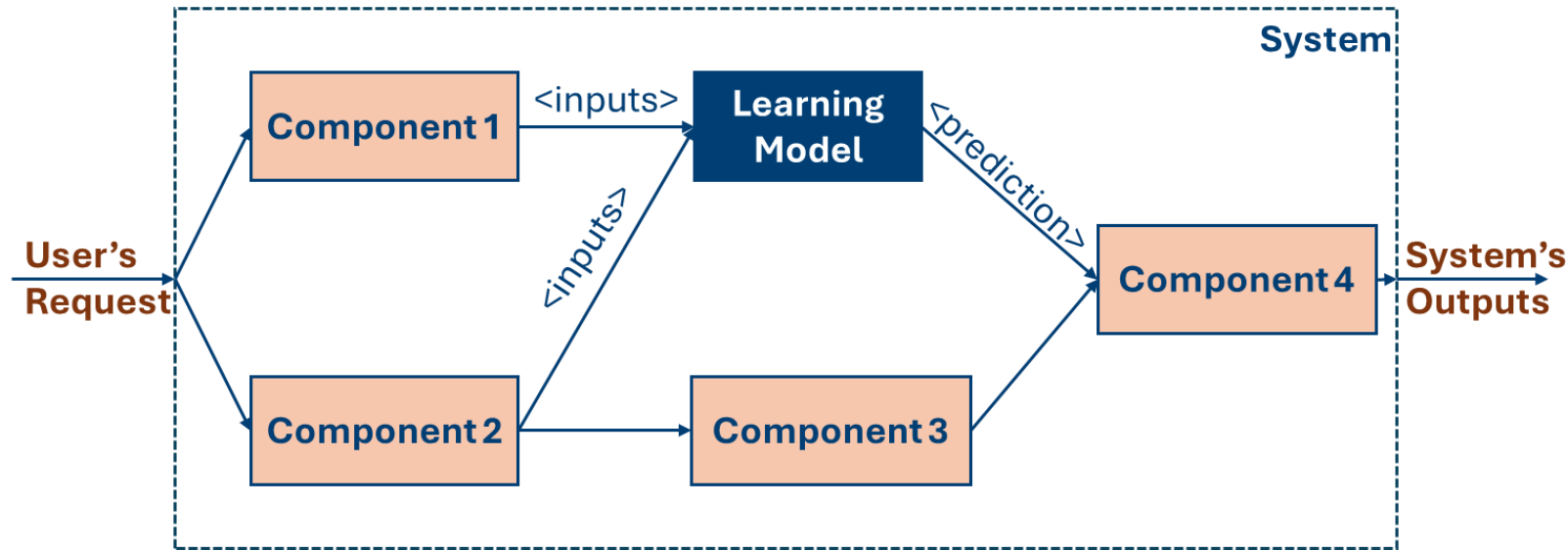


*** Low latency requirement! (< 5secs)**

Learning Model	Accuracy	Latency
Modelv1	88%	3 secs
Modelv2	90%	4 secs
Modelv3	98%	10 secs

Real-world Deployments

- Context matters



*** Low latency requirement! (< 5secs)**

*** Constraint resources.**

Learning Model	Accuracy	Latency	Resources Demand
Modelv1	88%	3 secs	Low
Modelv2	90%	4 secs	Medium
Modelv3	98%	10 secs	High

Context



Source:
https://commons.wikimedia.org/wiki/File:NP_coffee_cooperative_%285867722870%29.jpg

**What do people need?
What are the social problems
data science can help with?**



Source:
<https://www.flickr.com/photos/scottishgovernment/23657582298/in/p-hotostream/>

Technology Adoption



**Socio-Technical
Systems**

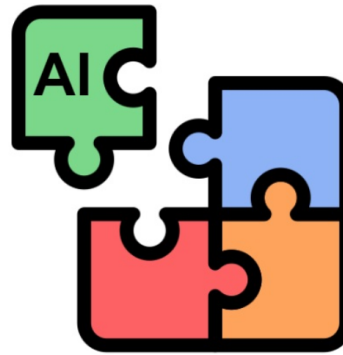


**Artificial
Intelligence**

Technology Adoption



**Socio-Technical
Systems**



**Software
Systems**



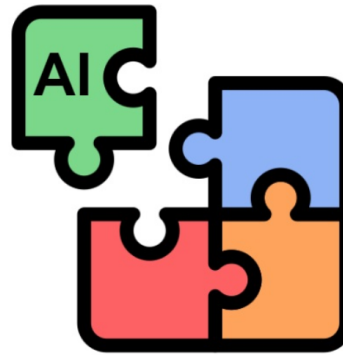
**Artificial
Intelligence**

Technology Adoption

PURPOSE



**Socio-Technical
Systems**



**Software
Systems**



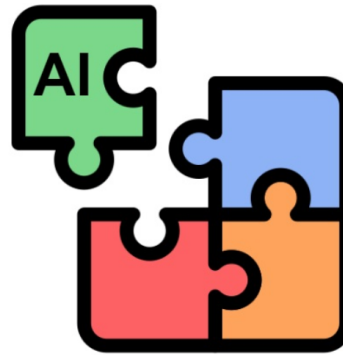
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PURPOSE



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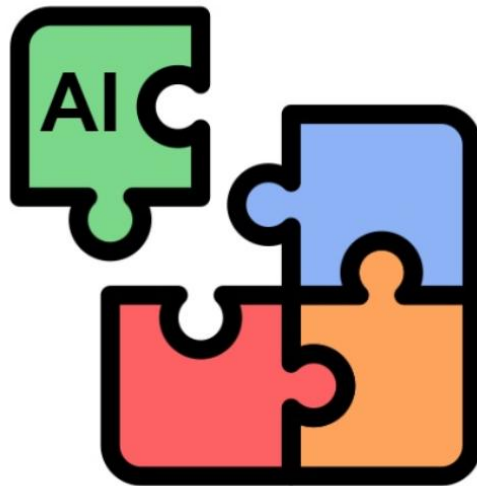
**Software
Systems**



**Artificial
Intelligence**

A D O P T I O N

How do we engineer software systems?

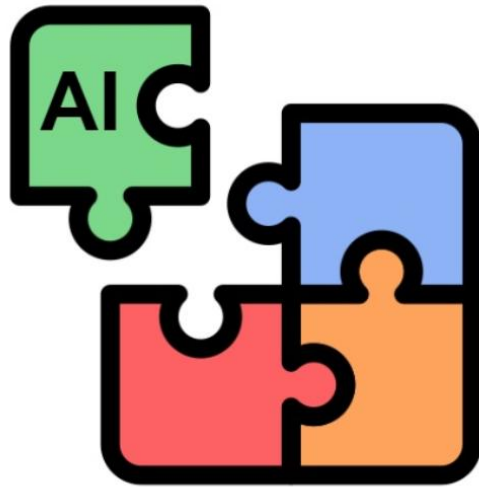


**Software
Systems**

The Systems Engineering Approach

- Problem first
 - Why is this problem important?
 - What are the people needs?
 - Which are the problem constraints?
 - What are the important variables to consider?
- ...

How do we engineer software systems?

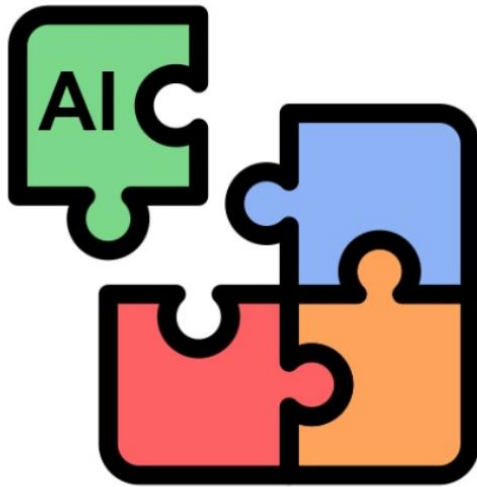


**Software
Systems**

The Systems Engineering Approach

- Engineering Principles
 - Systems Thinking
- Process Model

How do we engineer software systems?

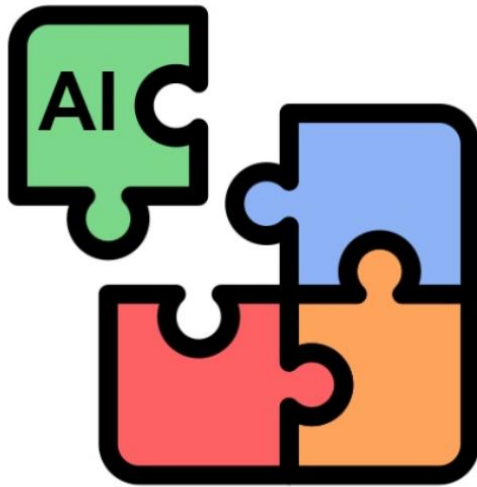


**Software
Systems**

Systems Thinking

- System Views: Defining problems from different perspectives.
- Agility System: Defining flexible systems architectures.
- System Dynamics: Modelling the changing nature of systems.

How do we engineer software systems?



**Software
Systems**

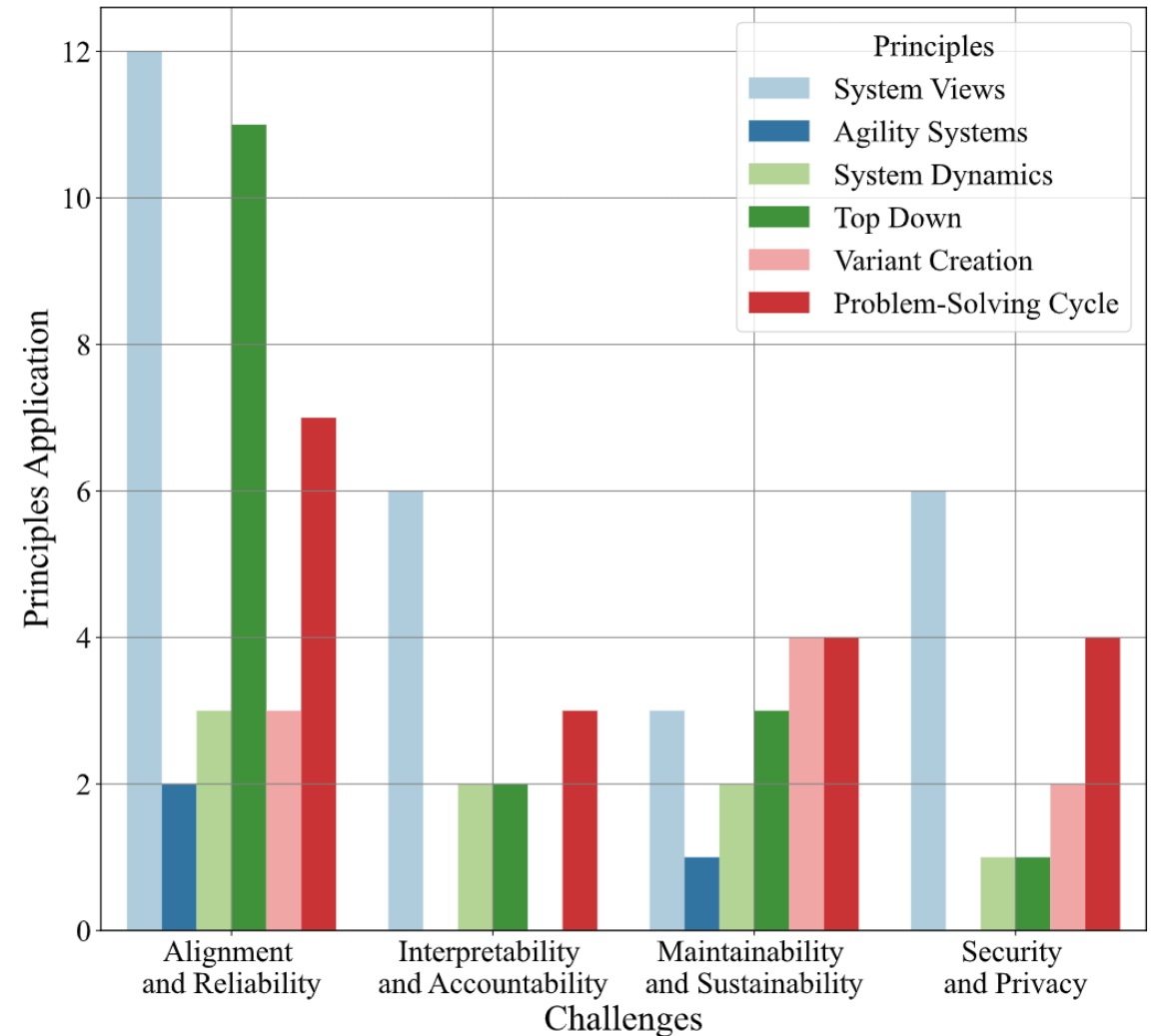
Process Model

- Top-Down Analysis: Divide and conquer. Problem decomposition.
- Variant Creation: Assessing different alternatives to solve a problem.
- Problem-Solving Cycle: Defining and following a methodology.

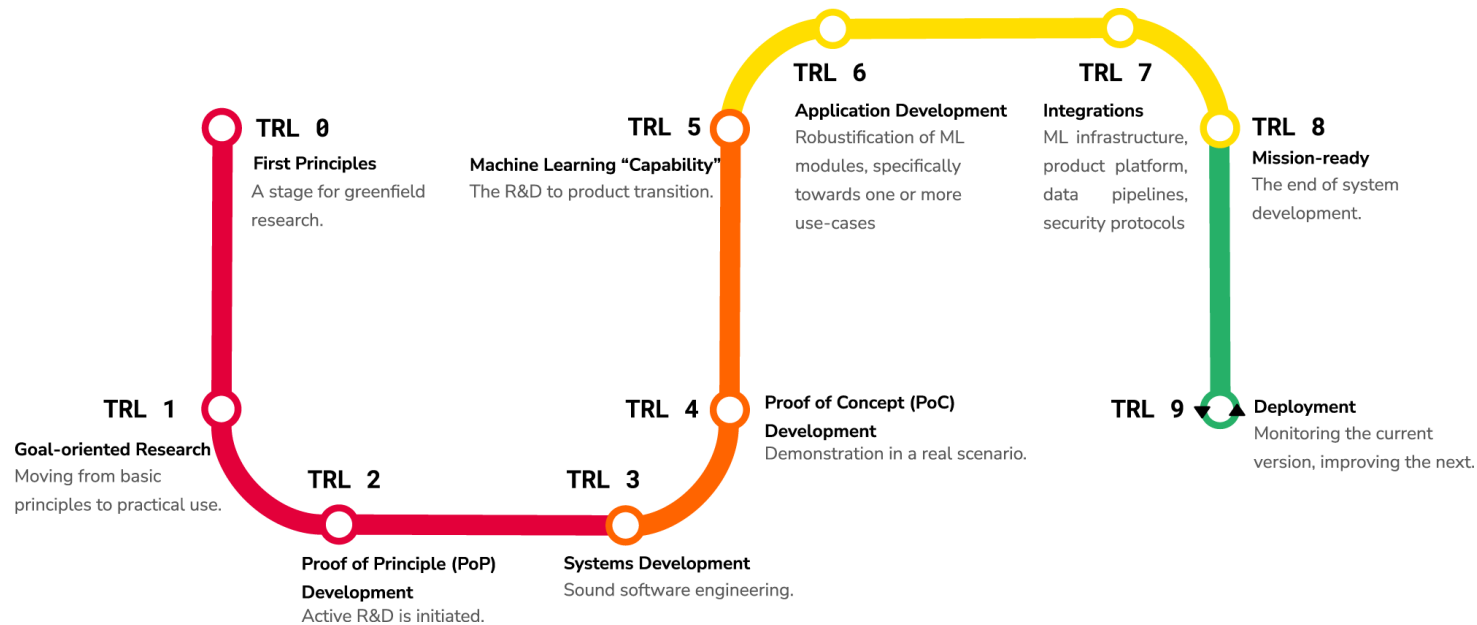
Systems Engineering Principles in Practice

A survey of 24 works that apply the principles to address when deploying AI-based Systems:

- Alignment and Reliability
- Interpretability and Accountability
- Maintainability and Sustainability
- Security and Privacy



Systems Engineering Principles in Practice

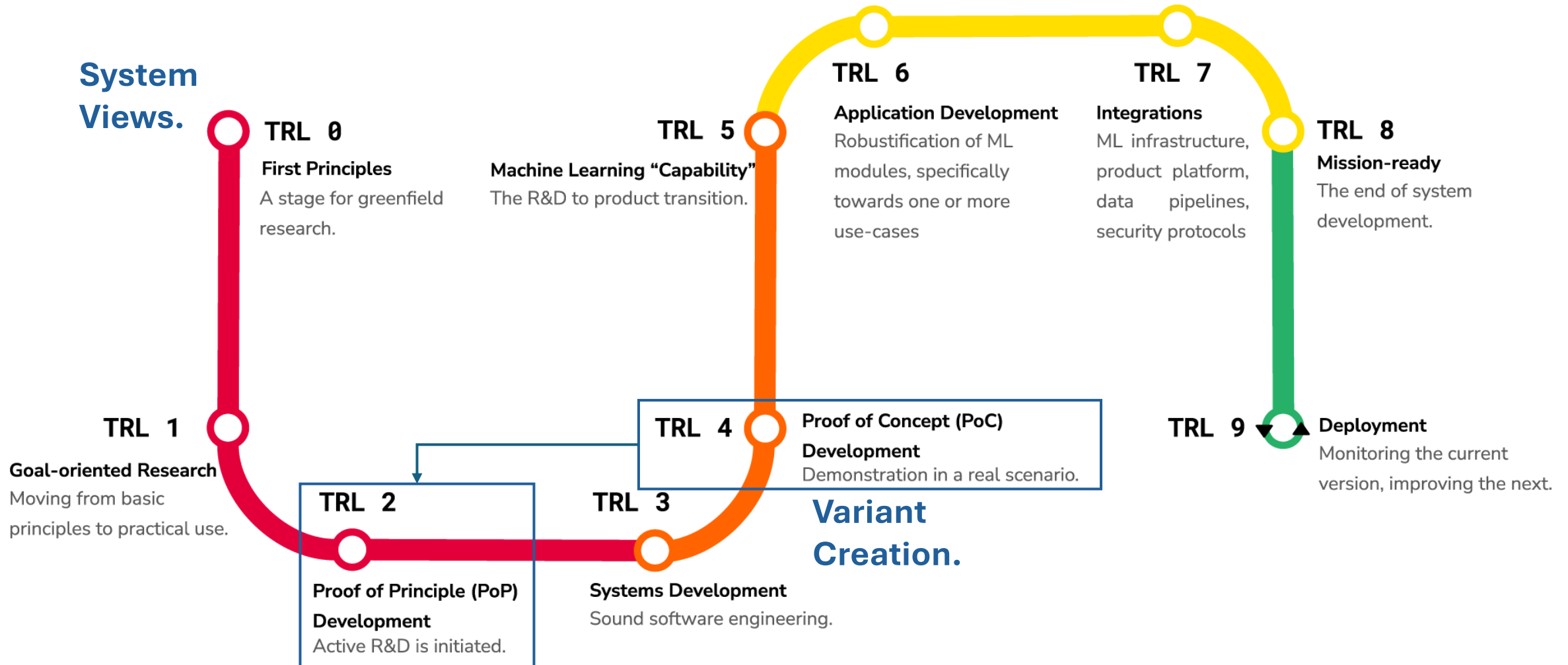


MLTRL – Technology readiness levels for machine learning systems

Domain: Critical Systems

Alignment and reliability, interpretability and accountability, maintainability and sustainability, and security and privacy.

Systems Engineering Principles in Practice



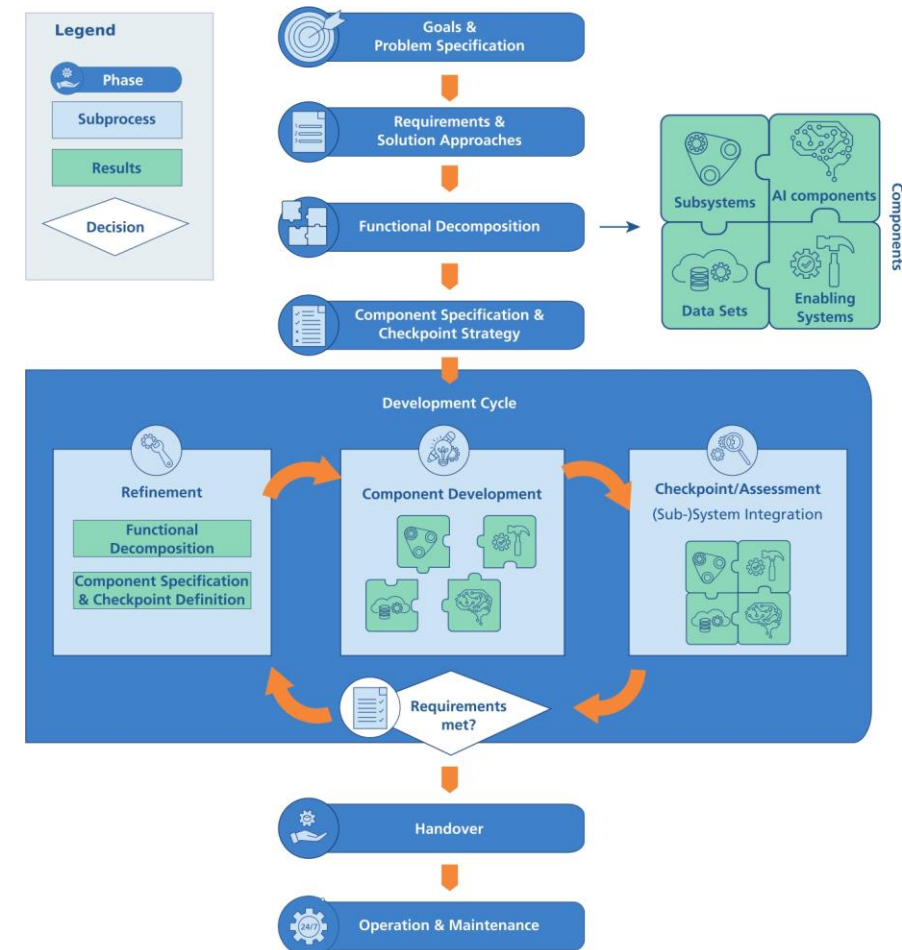
Systems Engineering Principles in Practice

PAISE® – Process Model for AI Systems Engineering

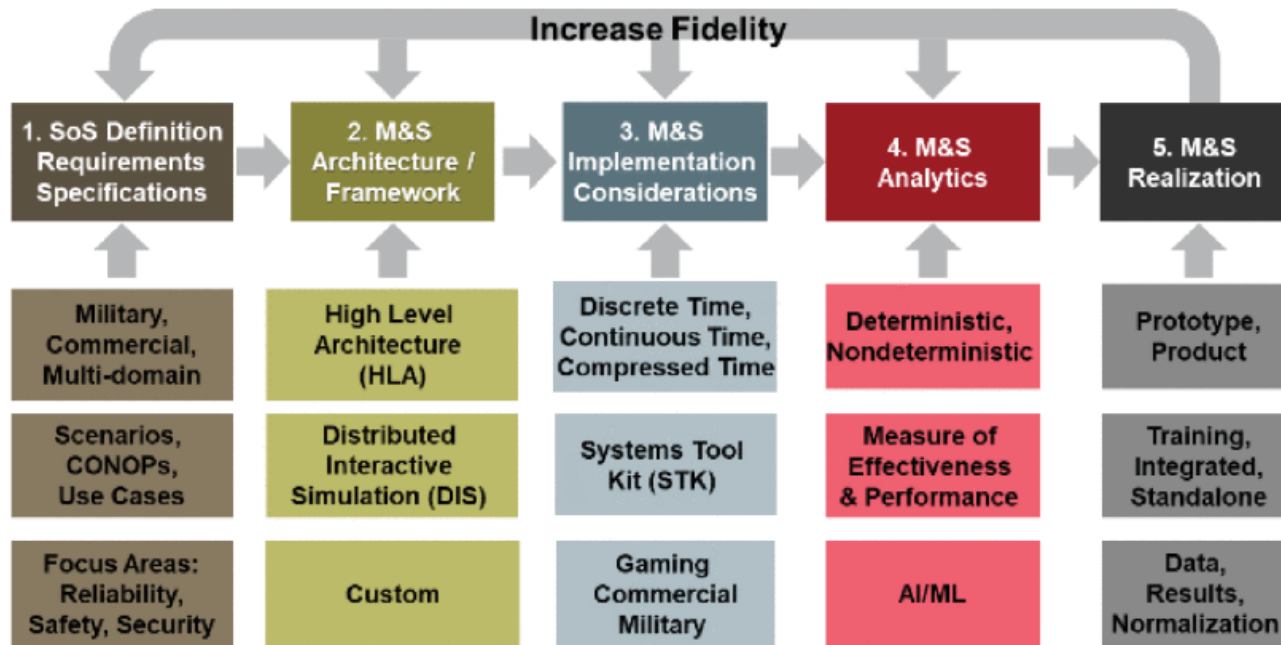
Domain: ML-based Systems

Alignment and reliability:

- Problem Solving Cycle.
- Top-Down Analysis.
- System Views.
- Agility Systems.



Systems Engineering Principles in Practice



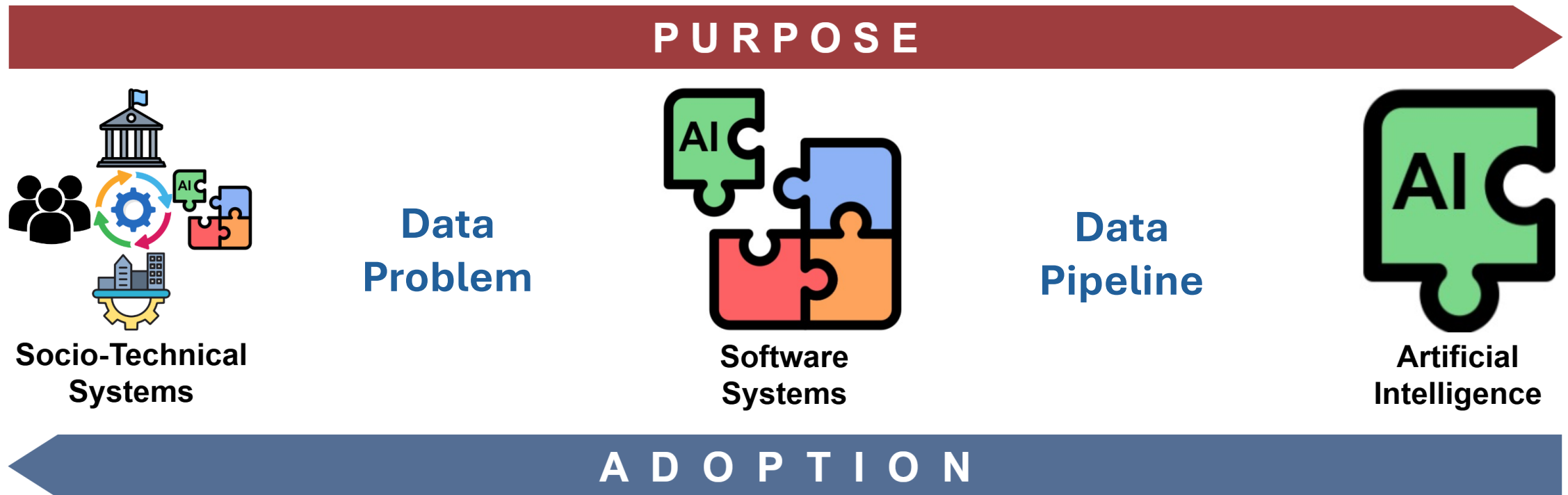
Alignment and reliability, maintainability and sustainability, and security and privacy.

ACDANS – System of Systems Engineering Approach for Complex Deterministic and Nondeterministic Systems

Domain: Military Systems

- Problem Solving Cycle.
- System Dynamics
- Variant Creation.

Engineering Data Science



Data Problem

- Understanding stakeholders' needs
- Understanding resources and constraints
- Understanding data sources
- Understanding the nature of the data

Data Problem

- Understanding stakeholders' needs
- Understanding resources and constraints
- Understanding data sources
- Understanding the nature of the data

Systems Thinking

- Different points of view: **city council, landlord, tenants.**
- Agility Systems: **Flexible resources (e.g., cloud database), and flexible architecture (i.e., library).**
- Dynamic Systems: **Heterogeneous data, temporal features, spatial data, etc.**

Data Pipeline

The Fynesse framework:

- Access
- Assess
- Address

Data Pipeline

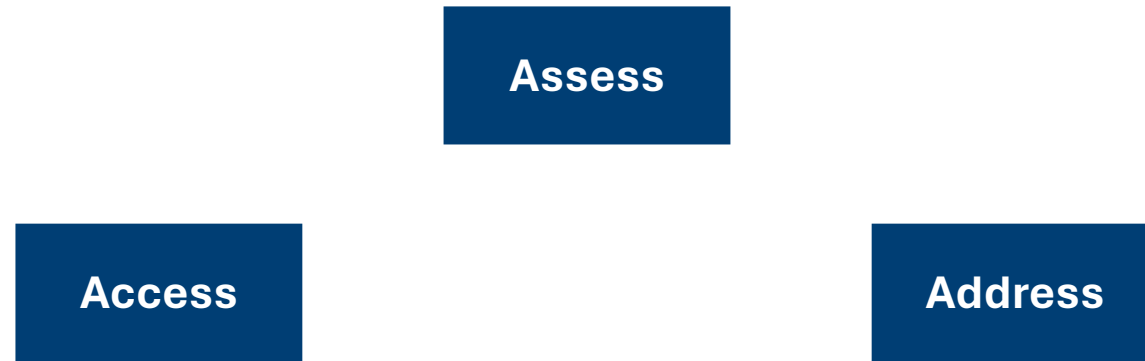
The Fynesse framework:

- Access
- Assess
- Address

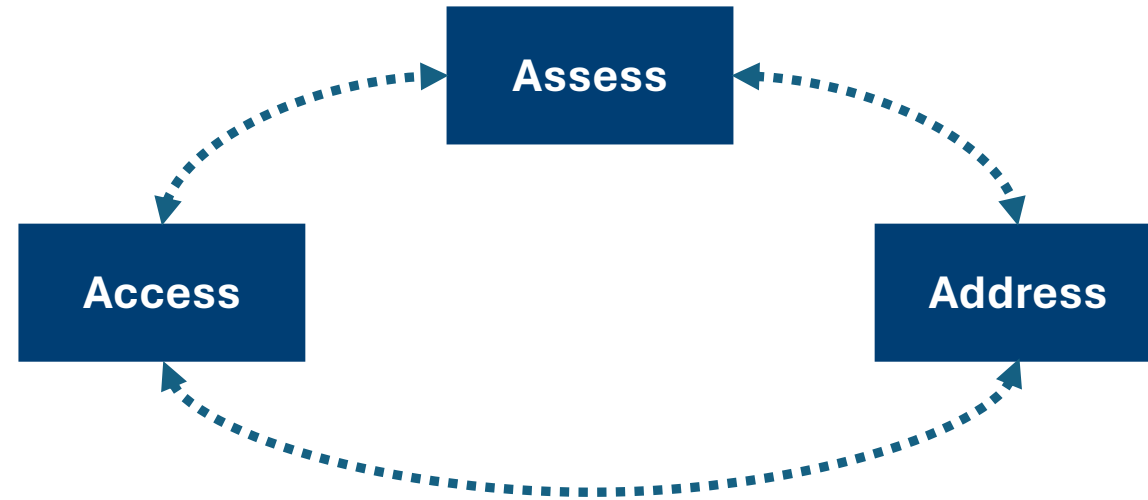
Process Model

- 
- Top-down: **downloading and uploading data chunks.**
 - Variant creation: **exploring different data sources and management strategies.**
 - Problem-solving cycle: **access, assess, and address loop.**

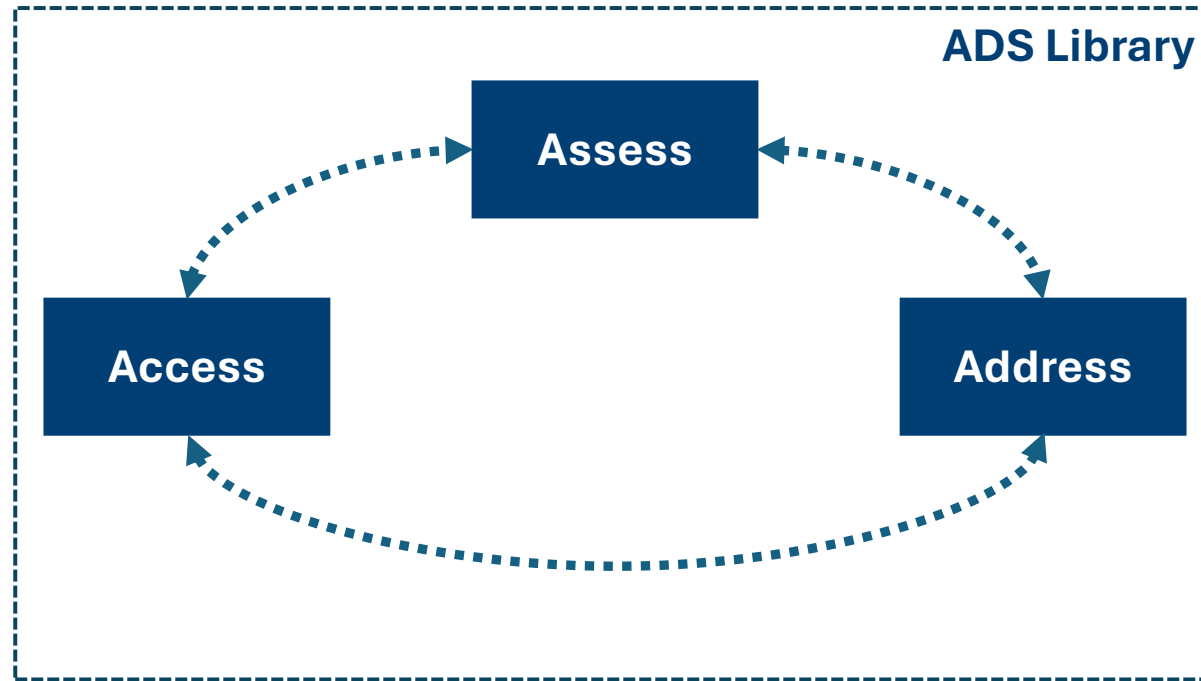
Engineering Data Science



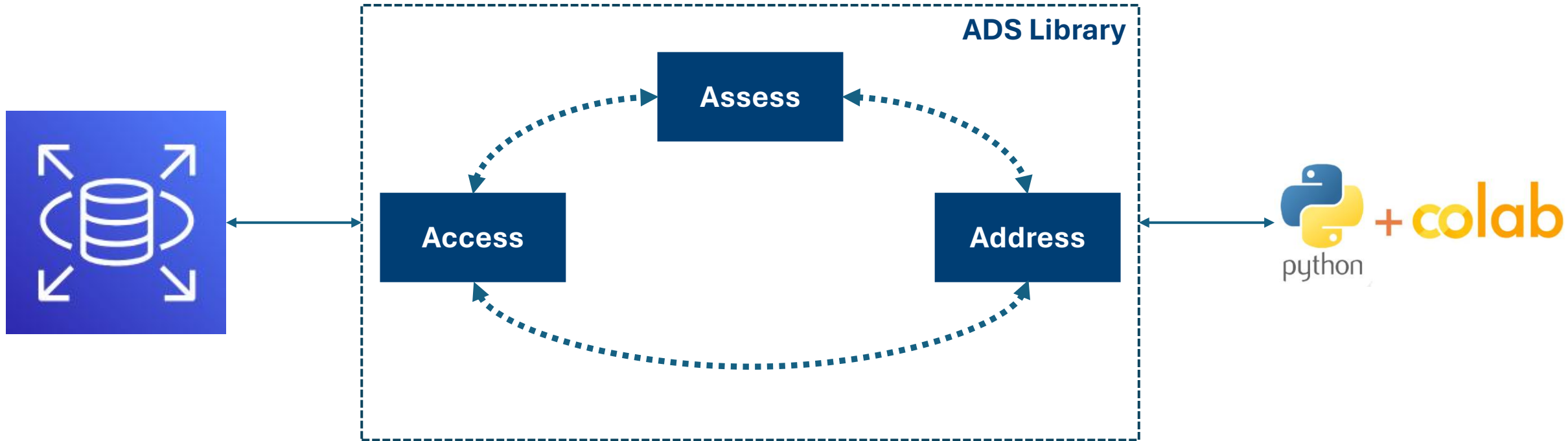
Engineering Data Science



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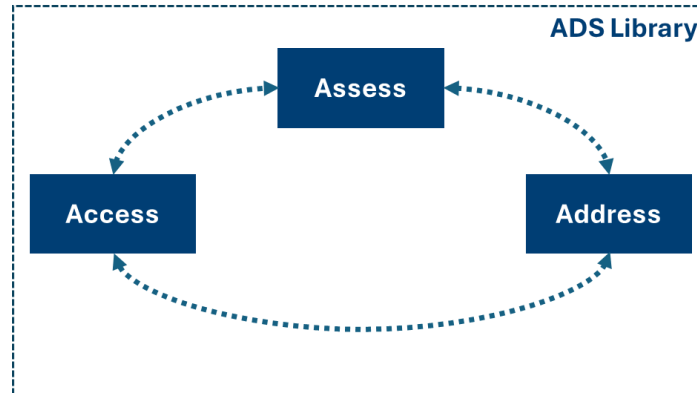
Engineering Data Science

- Data from UK citizens and infrastructure.
- gov.UK and institutions managing data.
- Open Street Maps (OSM).

PURPOSE



Socio-Technical
Systems



Artificial
Intelligence

ADOPTION

Learning models

Summary

- Context is important when addressing Data Science challenges.
- We must address these challenges because Data science projects impact people.
- The Systems Engineering approach provides principles to guide our work.
- We should include these principles in our Data Science projects.

Many thanks!